SMC Pneumatics System Manual - Modular FRL and Solenoid Manifolds

Air Preparation Assembly: SMC AC40-D modular filter, regulator, and lubricator unit.

Inlet Air Specification: Industrial plant supply 7.0 to 8.5 bar, dry non-lubricated air.

Nominal Cell Operating Pressure: Regulated to 6.2 bar secondary pressure.

Minimum Operating Pressure Threshold: 5.5 bar. If pressure falls below 5.5 bar, ALM-102 trips.

Safety Soft-Start Dump Valve: SMC VP517 connected to safety emergency stop contactor loop.

SMC SY5000 Series Valve Manifold Maintenance & Diagnostics

Valve Island: 8-station compact cassette manifold with integrated PROFINET communication bus.

Station 1 & 2: Part clamp and escapement cylinders on infeed conveyor CONV01.

Station 3 & 4: Gripper cylinder actuate and retract lines feeding ROBOT01 end effector.

Spool Sticking Issues: Contamination or moisture ingress causes sluggish spool travel or valve sticking.

Manifold Air Leaks: Inspect sub-base gasket O-rings and 8mm push-in fittings using ultrasonic leak detector.

SMC Pneumatics - Filtration, Condensate Drainage, and Pressure Switches

Coalescing Filter Maintenance: Replace 5-micron element when differential pressure indicator shows red.

Auto Drain Bowl: Regularly inspect float drain mechanism; purge manual drain valve monthly.

Electronic Pressure Switch: SMC ISE40A provides digital PNEUMATIC_OK signal to PLC01 input I0.7.

Adjust switch setpoint to 5.8 bar with 0.3 bar hysteresis to prevent nuisance tripping of alarm ALM-102.